

Week 6 Assignment: Git, GitHub & Final Data Analytics Project

Project Title: Olist E-Commerce Sales & Order Performance Analysis.

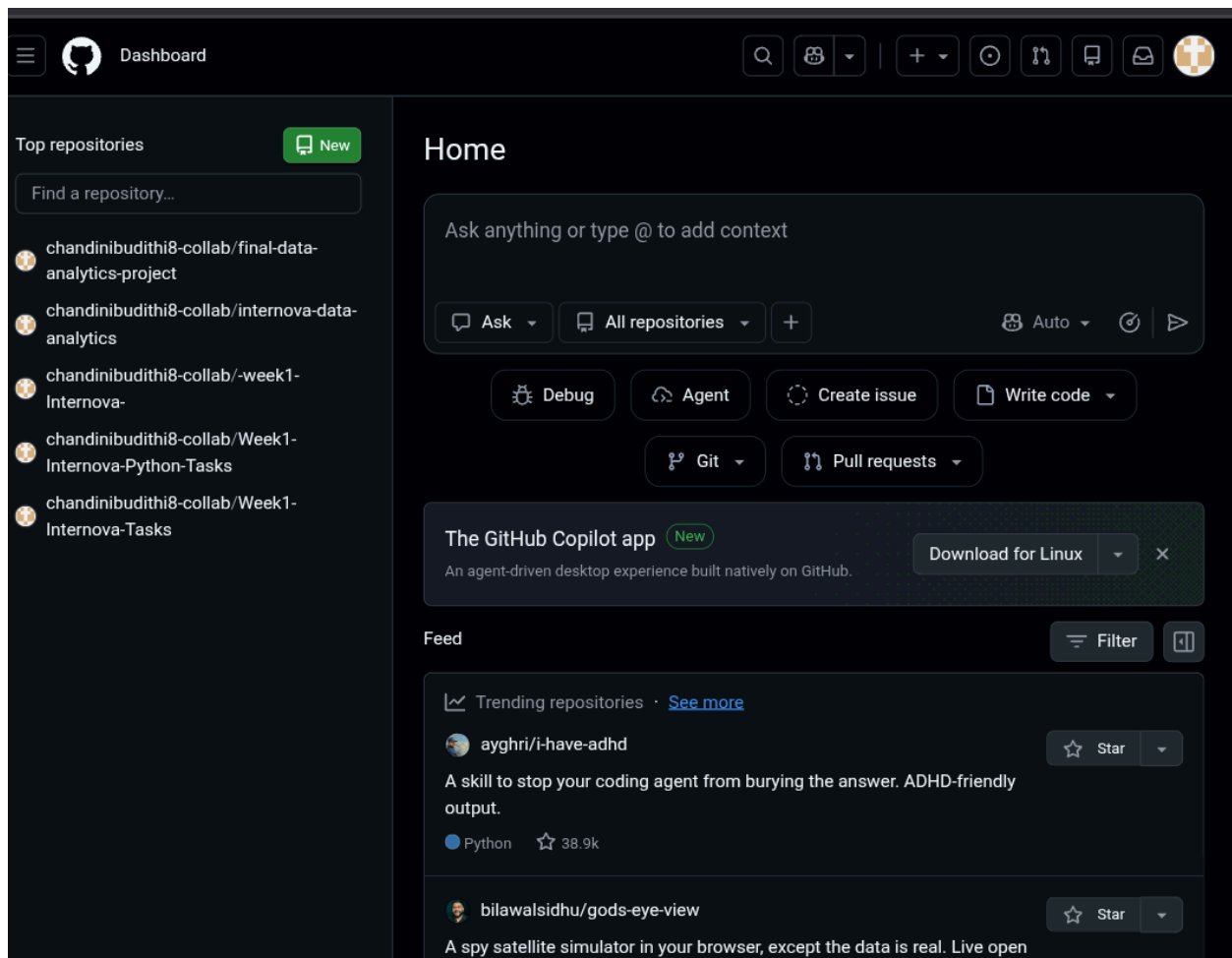
Task 1: Git & GitHub Basics

1) Install Git:

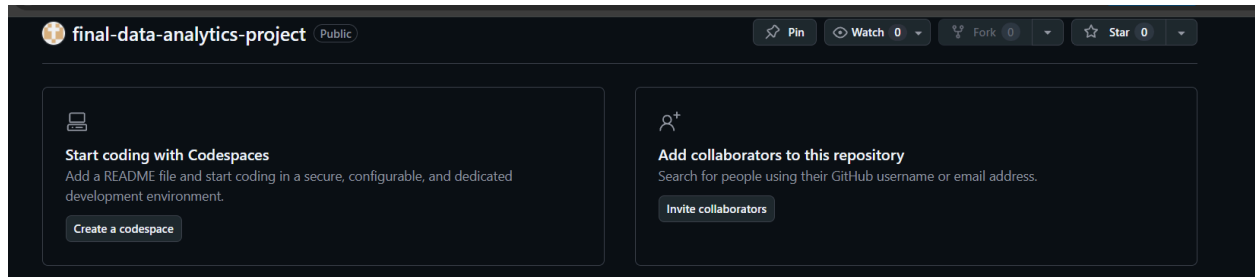
```
MINGW64:/c/Users/budithi chandini

budithi chandini@chandinibudithi MINGW64 ~
$ git --version
git version 2.55.0.windows.5
```

2) GitHub Account:



3) Repository Created:



4) Repository Cloned:

```
budithi chandini@chandinibudithi MINGW64 ~/Documents
$ </> Bash git clone https://github.com/chandinibudithi8-collab/final-data-analytics-project.git
Cloning into 'final-data-analytics-project'...
warning: You appear to have cloned an empty repository.
```

5)File Added:

```
budithi chandini@chandinibudithi MINGW64 ~/documents/final-data-analytics-project (main)
$ git add practice.txt
warning: in the working copy of 'practice.txt', LF will be replaced by CRLF the next time Git touches it

budithi chandini@chandinibudithi MINGW64 ~/documents/final-data-analytics-project (main)
$ git status
On branch main

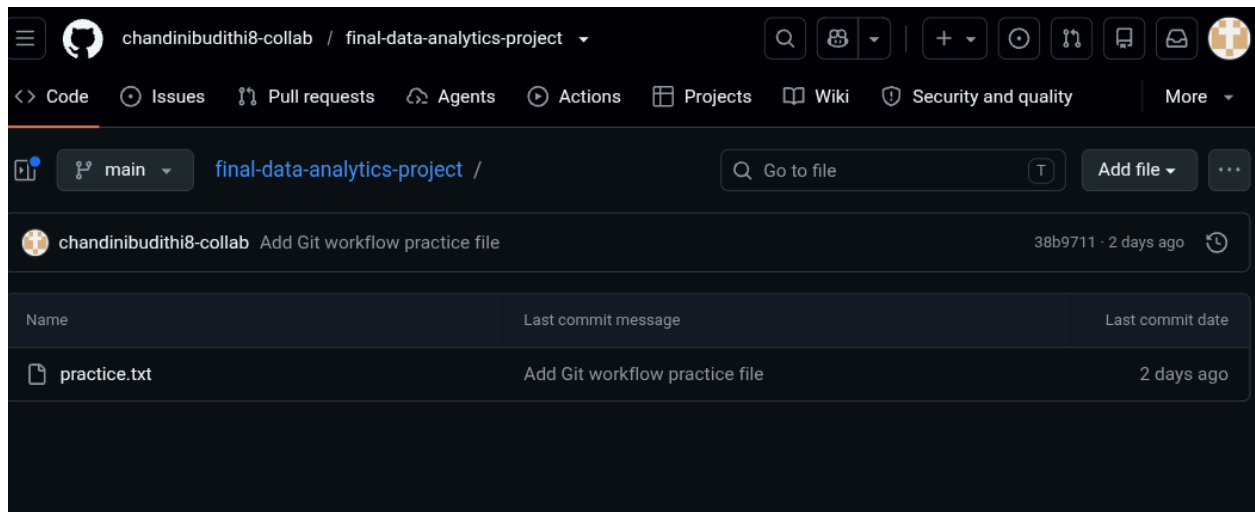
No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   practice.txt
```

6)Commit Created:

```
$ git commit -m "Add Git workflow practice file"
[main (root-commit) 38b9711] Add Git workflow practice file
 1 file changed, 1 insertion(+)
 create mode 100644 practice.txt
```

7)Push To GitHub:



Task 2: Git Workflow Practice

1)git clone:

```
budithi chandini@chandinibudithi MINGW64 ~/Documents
$ </> Bash git clone https://github.com/chandinibudithi8-collab/final-data-analytics-project.git
Cloning into 'final-data-analytics-project'...
warning: You appear to have cloned an empty repository.
```

Explanation:

The **git clone** command was used to create a local copy of the GitHub repository on the computer. This allowed the project files to be managed locally using Git and synchronized with the remote GitHub repository.

2)Git add:

```
budithi chandini@chandinibudithi MINGW64 ~/documents/final-data-analytics-project (main)
$ git add practice.txt
warning: in the working copy of 'practice.txt', LF will be replaced by CRLF the next time Git touches it

budithi chandini@chandinibudithi MINGW64 ~/documents/final-data-analytics-project (main)
$ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   practice.txt
```

Explanation:

The **git add** command was used to add the required project files to the Git staging area before creating a commit.

3)git commit:

```
budithi chandini@chandinibudithi MINGW64 ~/documents/final-data-analytics-project (main)
$ git commit -m "Add Git workflow practice file"
[main (root-commit) 38b9711] Add Git workflow practice file
 1 file changed, 1 insertion(+)
 create mode 100644 practice.txt
```

Command:git commit -m

Explanation:

The **git commit** command was used to record the staged project changes in the local Git repository. Meaningful commit messages were provided to describe the purpose of each change. Multiple commits were created while developing the final data analytics project.

4)git push:

```
budithi chandini@chandinibudithi MINGW64 ~/documents/final-data-analytics-project (main)
$ git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 259 bytes | 259.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/chandinibudithi8-collab/final-data-analytics-project.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Command: git push -u origin main

Explanation:

The git push command was used to upload the committed project changes from the local repository to the remote GitHub repository. This synchronized the local project with GitHub and made the committed files available in the online repository.

5)git pull:

```
budithi chandini@chandinibudithi MINGW64 ~/documents/final-data-analytics-project (main)
$ git pull origin main
From https://github.com/chandinibudithi8-collab/final-data-analytics-project
* branch      main      -> FETCH_HEAD
Already up to date.
```

Command:git pull

Explanation: **git pull** gets the latest changes from the GitHub repository (remote) and brings them into your local repository. Technically, it fetches the remote changes and then integrates them into your current branch.

6) 3 Commits:

Commit 1 — Add final data analytics project notebook

```
budithi chandini@chandinibudithi MINGW64 ~/final-data-analytics-project (main)
$ git add "Final_Data_Analytics_Project.ipynb"
warning: in the working copy of 'Final_Data_Analytics_Project.ipynb', LF will be replaced by CRLF the next time Git touches it

budithi chandini@chandinibudithi MINGW64 ~/final-data-analytics-project (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   Final_Data_Analytics_Project.ipynb

budithi chandini@chandinibudithi MINGW64 ~/final-data-analytics-project (main)
$ git commit -m "Add final data analytics project notebook"
[main 85a345b] Add final data analytics project notebook
 1 file changed, 2835 insertions(+)
 create mode 100644 Final_Data_Analytics_Project.ipynb
```

Explanation: This commit added the completed Python notebook containing the data analytics project work, including data preparation, exploratory data analysis, and analysis performed using Python.

Commit 2 — Add Power BI sales dashboard

```
budithi chandini@chandinibudithi MINGW64 ~/final-data-analytics-project (main)
$ git add "olist_ecommerce _dashboard.pbix"

budithi chandini@chandinibudithi MINGW64 ~/final-data-analytics-project (main)
$ git commit -m "Add Power BI sales dashboard"
[main d6213d6] Add Power BI sales dashboard
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 olist_ecommerce _dashboard.pbix
```

Explanation: This commit added the Power BI dashboard created for the Olist e-commerce project. The dashboard presents key sales and order performance metrics and visualizations.

Commit 3 — Add project visualizations

This commit added the project

```
budithi chandini@chandinibudithi MINGW64 ~/final-data-analytics-project (main)
$ git add "visualisations"

budithi chandini@chandinibudithi MINGW64 ~/final-data-analytics-project (main)
$ git commit -m "Add Project Visualisations"
[main cfa34f7] Add Project Visualisations
16 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 visualisations/Screenshot 2026-09-10 193731.png
create mode 100644 visualisations/Screenshot 2026-09-10 193934.png
create mode 100644 visualisations/Screenshot 2026-09-10 194209.png
create mode 100644 visualisations/Screenshot 2026-09-10 204003.png
create mode 100644 visualisations/Screenshot 2026-09-10 204349.png
create mode 100644 visualisations/Screenshot 2026-09-10 211411.png
create mode 100644 visualisations/Screenshot 2026-09-10 212458.png
create mode 100644 visualisations/Screenshot 2026-09-10 212933.png
create mode 100644 visualisations/Screenshot 2026-09-10 213157.png
create mode 100644 visualisations/Screenshot 2026-09-10 213803.png
create mode 100644 visualisations/Screenshot 2026-09-10 213840.png
create mode 100644 visualisations/Screenshot 2026-09-10 215228.png
create mode 100644 visualisations/Screenshot_2026_0910_165638 - Copy.png
create mode 100644 visualisations/Screenshot_2026_0910_172330.png
create mode 100644 visualisations/Screenshot_2026_0910_173607.png
create mode 100644 visualisations/Screenshot_2026_0910_181342.png
```

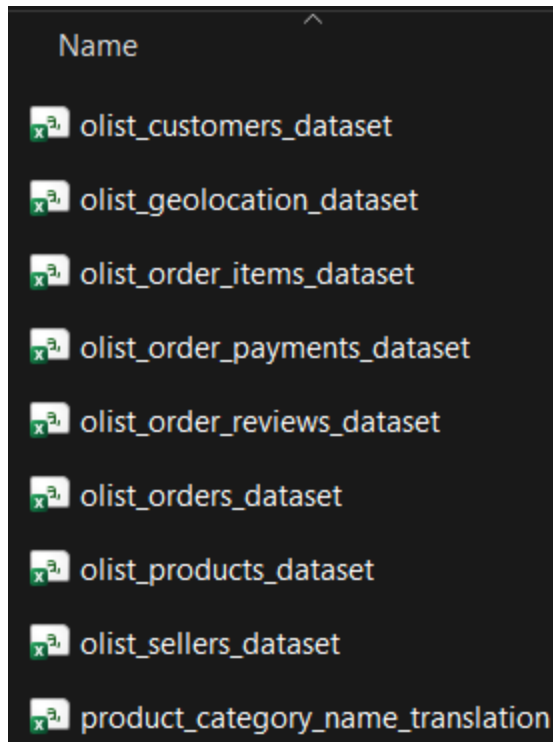
Explanation: visualization files, including the Python analysis charts and the Power BI dashboard screenshot. These visualizations provide graphical representation of the project findings.

Conclusion:

These three meaningful commits demonstrate how Git was used to track different stages of the final data analytics project. Each commit records a specific set of project changes in the Git history.

Task 3: Dataset Selection & Data Preparation:

1)Data Selection:Olist Brazilian E-Commerce Public Dataset



Explanation: The Brazilian E-Commerce Public Dataset by Olist was selected for this project. It is a real-world e-commerce dataset containing approximately 100,000 orders made between 2016 and 2018. The dataset provides information about orders, customers, products, sellers, payments, reviews, and delivery performance, making it suitable for data analysis and business insights.

2)Data Inspection:

import pandas as pd

orders=pd.read_csv('olist_orders_dataset.csv')

Code:orders.head()

	order_id	customer_id	order_status	order_purchase
0	e481f51cbdc54678b7cc49136f2d6af7	9ef432eb6251297304e76186b10a928d	delivered	2017-10
1	53cdb2fc8bc7dce0b6741e2150273451	b0830fb4747a6c6d20dea0b8c802d7ef	delivered	2018-01
2	47770eb9100c2d0c44946d9cf07ec65d	41ce2a54c0b03bf3443c3d931a367089	delivered	2018-01
3	949d5b44dbf5de918fe9c16f97b45f8a	f88197465ea7920adcdbec7375364d82	delivered	2017-11
4	ad21c59c0840e6cb83a9ceb5573f8159	8ab97904e6daea8866dbdbc4fb7aad2c	delivered	2018-01

orders.info()

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99441 entries, 0 to 99440
Data columns (total 8 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   order_id                             99441 non-null  object
1   customer_id                         99441 non-null  object
2   order_status                         99441 non-null  object
3   order_purchase_timestamp            99441 non-null  object
4   order_approved_at                   99281 non-null  object
5   order_delivered_carrier_date        97658 non-null  object
6   order_delivered_customer_date       96476 non-null  object
7   order_estimated_delivery_date       99441 non-null  object
dtypes: object(8)
memory usage: 6.1+ MB

```

Explanation: The dataset was inspected using Pandas head() and info(). The head() method was used to examine the first few records, while info() was used to understand the dataset structure, column names, data types, and non-null values.

3) Missing Values:

Code: orders.isnull().sum()

	0
order_id	0
customer_id	0
order_status	0
order_purchase_timestamp	0
order_approved_at	160
order_delivered_carrier_date	1783
order_delivered_customer_date	2965
order_estimated_delivery_date	0
dtype: int64	

Explanation: The Orders table contains some missing values in the order approval and delivery date columns. These missing values occur because some orders did not reach the corresponding stage of the order process. Therefore, the missing dates were kept as they were instead of adding artificial dates, so that the analysis would remain accurate.

4)Handling Missing Values:

Missing delivery-related dates were retained as NaT because they represent order stages that had not occurred. Filling these dates with artificial values could give incorrect delivery-time results.

5)Duplicate:

Code:orders.duplicated().sum()

```
np.int64(0)
```

6)prepare the Orders table:

After inspecting and cleaning the Order Table the data was prepared for analysis. The relevant CSV files were loaded into Pandas and the required columns were retained for analysis. Date columns in the Orders table were converted into proper datetime format so that order trends and delivery-related analysis could be performed correctly. The missing delivery-related dates were retained as missing values because they represent orders that had not reached the corresponding stage of the order process. Duplicate records were checked and unnecessary or incorrect records were removed where required. The prepared datasets were then used for Exploratory Data Analysis (EDA), data visualization, and the Power BI dashboard.

Task 4:Exploratory Data Analysis

1)Descriptive statistics:

Code:order_items[['price','freight_value']].describe()

	price	freight_value
count	112650.000000	112650.000000
mean	120.653739	19.990320
std	183.633928	15.806405
min	0.850000	0.000000
25%	39.900000	13.080000
50%	74.990000	16.260000
75%	134.900000	21.150000
max	6735.000000	409.680000

Explanation: Descriptive statistics were calculated to understand the numerical characteristics of the dataset. Measures such as mean, median, minimum, maximum, and quartiles were examined for important numerical variables such as product price and freight value.

Average product price: R\$120.65

Median product price: R\$74.99

Minimum price: R\$0.85

Maximum price: R\$6,735

Average freight value: approximately R\$19.99

Median freight value: approximately R\$16.26

Maximum freight value: approximately R\$409.68

2)Data Inspection:

Code: order_items.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 112650 entries, 0 to 112649
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   order_id              112650 non-null object
1   order_item_id         112650 non-null int64
2   product_id            112650 non-null object
3   seller_id             112650 non-null object
4   shipping_limit_date    112650 non-null object
5   price                 112650 non-null float64
6   freight_value         112650 non-null float64
dtypes: float64(2), int64(1), object(4)
memory usage: 6.0+ MB
```

Explanation: The order Table consists 112650 rows and the columns are 7 ,Here the There is no null values in the order items table .The table is Accurate.And the data types are float,int,object .But Here we will covert the Shipping limit date Object type to date then the results are in accurate.

3)Data Cleaning:

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 112650 entries, 0 to 112649
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   order_id              112650 non-null  object
1   order_item_id         112650 non-null  int64
2   product_id            112650 non-null  object
3   seller_id             112650 non-null  object
4   shipping_limit_date    112650 non-null  datetime64[ns]
5   price                 112650 non-null  float64
6   freight_value         112650 non-null  float64
dtypes: datetime64[ns](1), float64(2), int64(1), object(3)
memory usage: 6.0+ MB

```

Explanation: In Data Cleaning Here we converted the type conversion from object to datetime which is shipping limit date ,then we get accurate results while performing calculations.

4) Missing Values Handling

Code: `order_items.isnull().sum()` #Inspecting missing values

```
#Inspecting missing values  
order_items.isnull().sum()
```

0

order_id	0
----------	---

order_item_id	0
---------------	---

product_id	0
------------	---

seller_id	0
-----------	---

shipping_limit_date	0
---------------------	---

price	0
-------	---

freight_value	0
---------------	---

dtype: int64

Explanation:Missing values were checked using the Pandas isnull() method. The analysis showed that the Orders table did not contain any missing values. Therefore, no missing-value treatment was required for the Orders table.

#Check duplicate values:

Code:order_items.duplicated().sum()

```
np.int64(0)
```

Explanation:Duplicate records were checked using the Pandas duplicated() method. The result was 0, indicating that no duplicate rows were found in the dataset. Therefore, no duplicate records needed to be removed.

5) Correlation Analysis:

Code:order_items[['price','freight_value']].corr()

	price	freight_value
price	1.000000	0.414204
freight_value	0.414204	1.000000

Explanation:The correlation between product price and freight value is 0.414, indicating a moderate positive relationship. Higher-priced products tend to have higher freight costs, although other factors also influence shipping costs.

6)Identification of important patterns and trends:

Code:order_items['price'].describe()

price	
count	112650.000000
mean	120.653739
std	183.633928
min	0.850000
25%	39.900000
50%	74.990000
75%	134.900000
max	6735.000000
dtype: float64	

Explanation: The mean price (₹120.65) is higher than the median price (₹74.99), suggesting that the price distribution is right-skewed due to some relatively high-priced products.

7) Identification of outliers where applicable:

Code: `order_items['price'].quantile([0.25,0.75])`

price	
0.25	39.9
0.75	134.9
dtype: float64	

```
Q1=order_items['price'].quantile(0.25)
```

```
Q3=order_items['price'].quantile(0.75)
```

```
IQR=Q3-Q1
```

```
print("IQR:",IQR)
```

```
print("Lower_Limit:",Q1-1.5*IQR)
```

```
print("Upper_Limit:",Q3+1.5*IQR)
```

```
IQR: 95.0
Lower_Limit: -102.6
Upper_Limit: 277.4
```

Explanation: The IQR method was used to identify potential outliers. The first quartile (Q1) was -102.6 and the third quartile (Q3) was 277.4, giving an IQR of 380. The lower and upper limits were calculated as -672.6 and 847.4 respectively. Values outside these limits were identified as potential outliers for further analysis.

8)Write a brief explanation of your findings:

EDA Findings

The dataset was explored using descriptive statistics, data inspection, data cleaning, missing-value analysis, correlation analysis, and outlier detection.

- The average product price was ₹120.65, while the median price was ₹74.99. The higher mean compared with the median suggests that the price distribution is right-skewed.
- There was a moderate positive correlation (0.414) between "price" and "freight_value", indicating that higher-priced items tend to have higher shipping costs.
- Using the IQR method, prices above ₹277.40 were identified as potential outliers. The maximum price of ₹6,735 confirms the presence of very high-priced items.
- These high-priced values were retained because they may represent genuine products rather than data-entry errors.
- Duplicate records were not found in the analyzed order-items data.

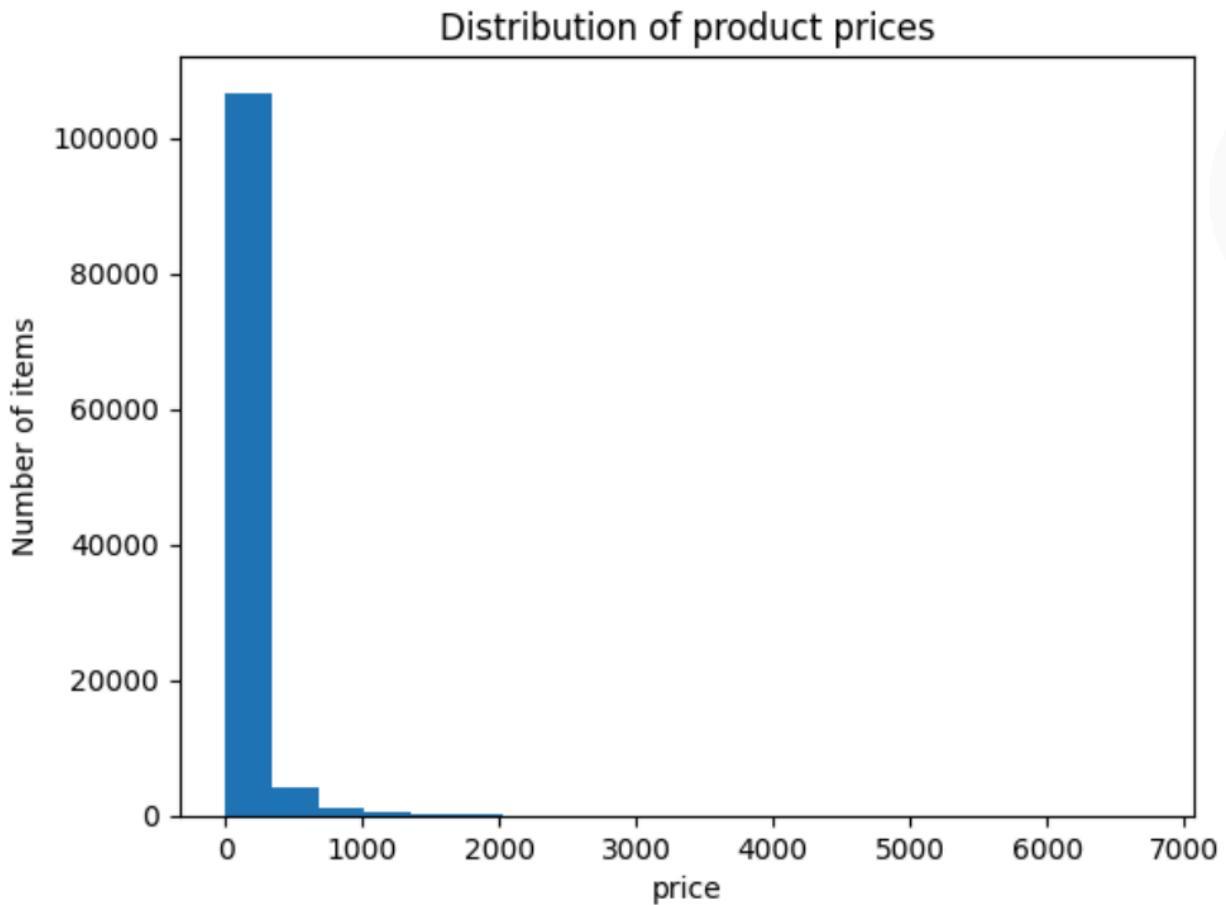
Task 5: Data Visualization

Visualisations:Python visualization libraries such as Matplotlib and Seaborn were used to create charts for understanding important patterns and relationships in the Olist e-commerce dataset. The visualizations helped present the results of the exploratory data analysis in an easy-to-understand graphical form.

1)price Distribution:

```
import matplotlib.pyplot as plt
plt.hist(order_items['price'],bins=20)
plt.title('Distribution of product prices')
```

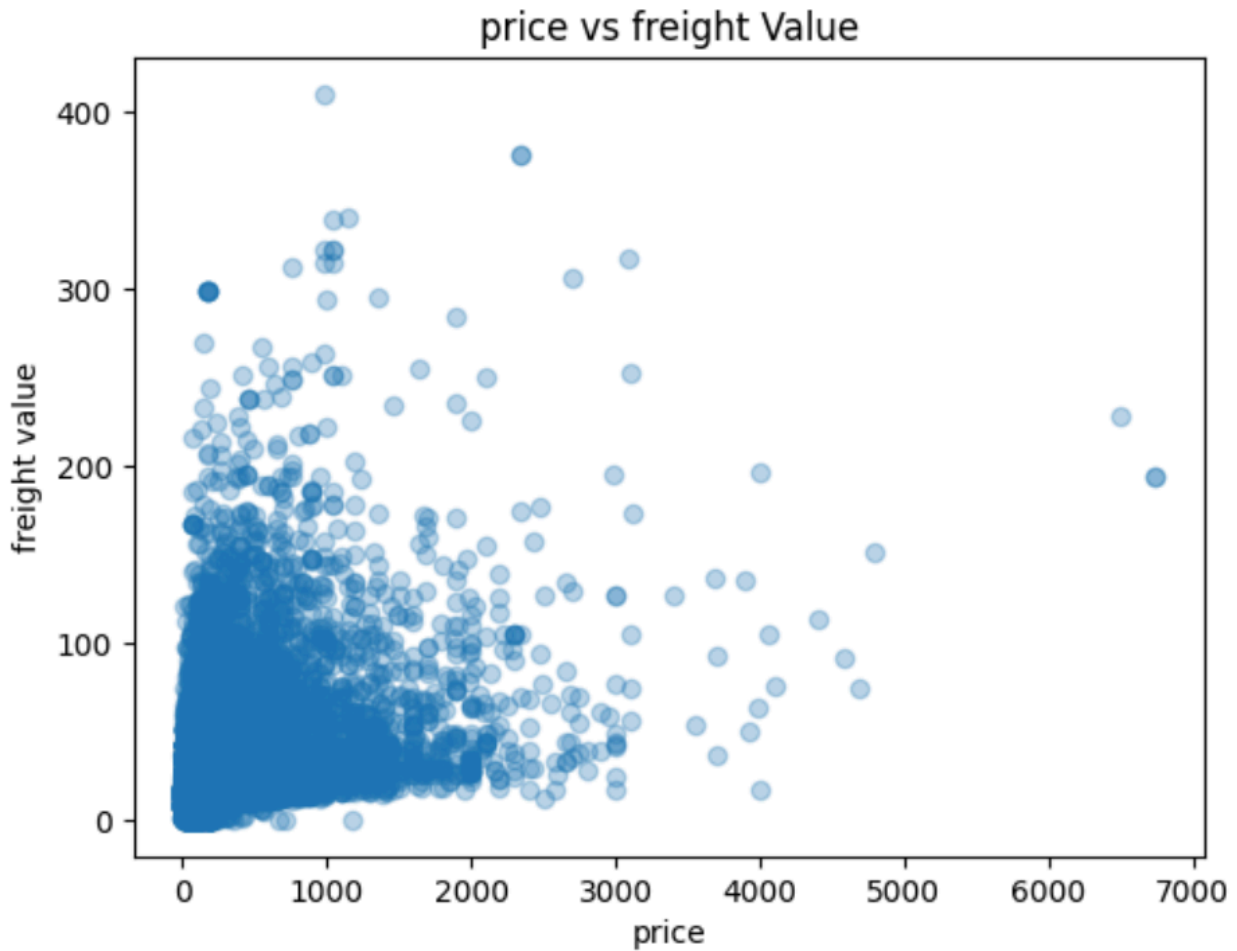
```
plt.xlabel('price')
plt.ylabel('Number of items')
plt.show()
```



Explanation: A histogram was used to analyze the distribution of product prices. The distribution is right-skewed, showing that most products have relatively lower prices, while a smaller number of products have considerably higher prices.

2) price vs Freight Value:

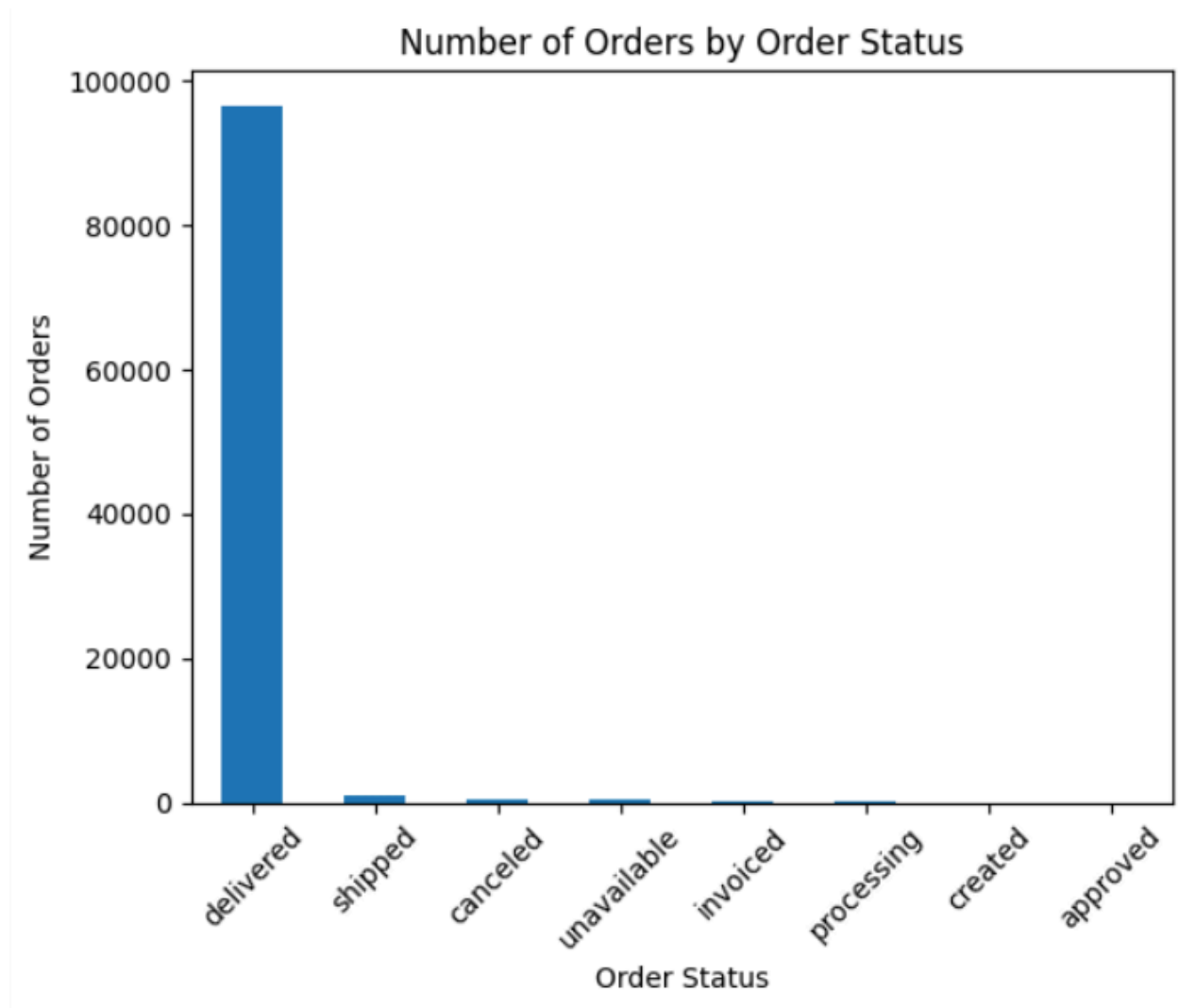
```
plt.scatter(order_items['price'],order_items['freight_value'],alpha=0.3)
plt.title("price vs freight Value")
plt.xlabel('price')
plt.ylabel('freight value')
plt.show()
```



Explanation: A scatter plot was used to analyze the relationship between product price and freight value. The plot shows a moderate positive relationship, with a correlation coefficient of approximately 0.414. This indicates that higher-priced products tend to have somewhat higher freight values, although the relationship is not very strong.

3)order status Bar:

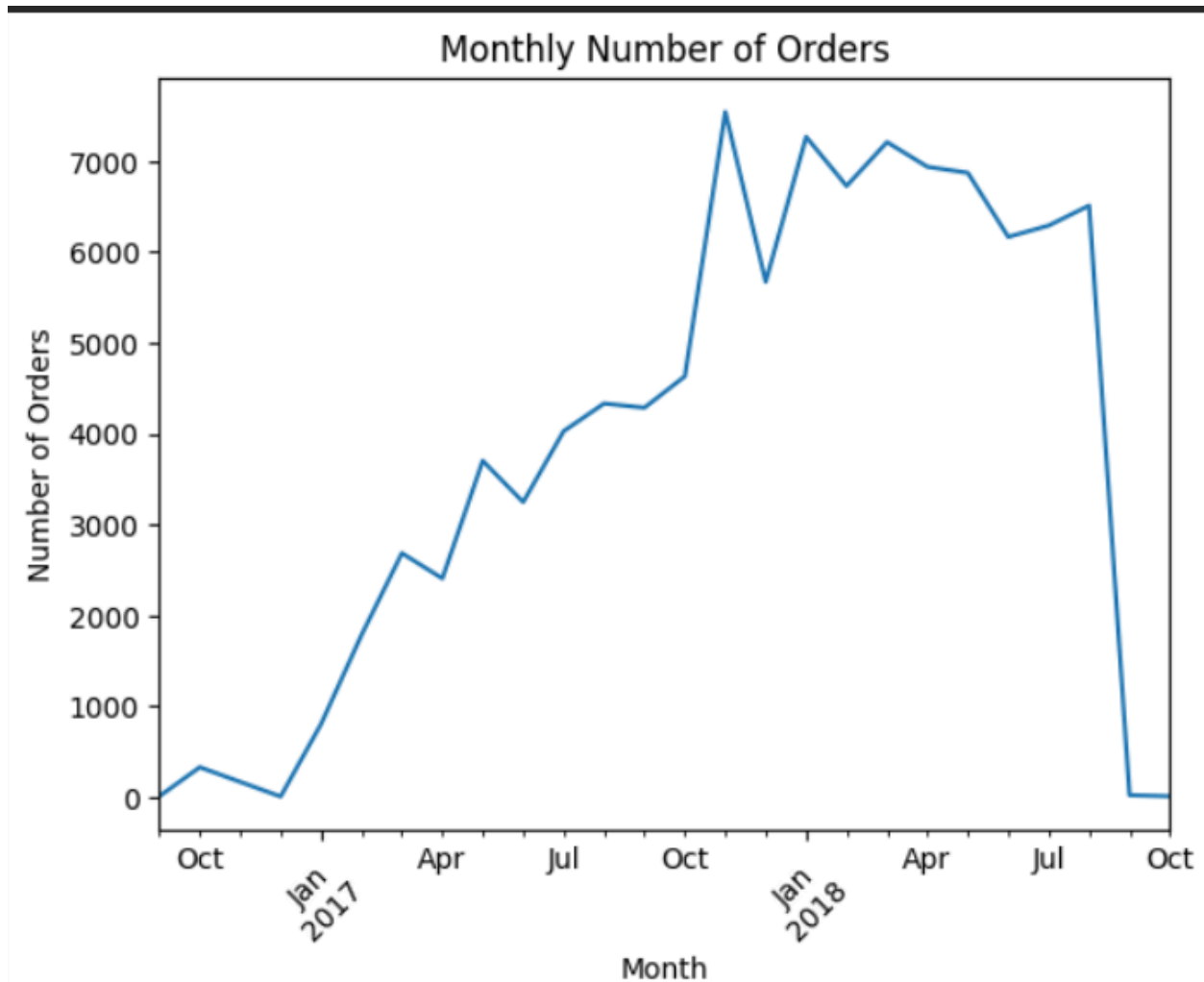
```
import matplotlib.pyplot as plt
orders_clean['order_status'].value_counts().plot(kind='bar')
plt.title('Number of Orders by Order Status')
plt.xlabel('Order Status')
plt.ylabel('Number of Orders')
plt.xticks(rotation=45)
plt.show()
```



Explanation: The bar chart shows the distribution of orders across different order statuses. Most orders were successfully delivered, while shipped, canceled, unavailable, and other statuses accounted for much smaller proportions. This indicates a high level of successful order fulfillment, while the smaller non-delivered categories may require further investigation.

4) Monthly Order Trends:

```
monthly_orders=orders['order_purchase_timestamp'].dt.to_period('M').value_counts().sort_index()
monthly_orders.plot(kind='line')
plt.title("Monthly Number of Orders")
plt.xlabel("Month")
plt.ylabel("Number of Orders")
plt.xticks(rotation=45)
plt.show()
```



Explanation: The line chart shows the monthly trend in the number of orders. Order volume was relatively lower during the earlier months and increased noticeably from around October 2017 onward. This indicates an overall growth in customer ordering activity during the later period of the dataset.

5) Top Product Categories:

Merge order items with products

```
category_orders=order_items.merge(  
products[['product_id','product_category_name']],on='product_id',how='left')
```

Count items sold by product category

```
category_counts=category_orders['product_category_name'].value_counts()
```

#10 categories with the highest number of items sold.

```
top_categories = category_counts.head(10)
```

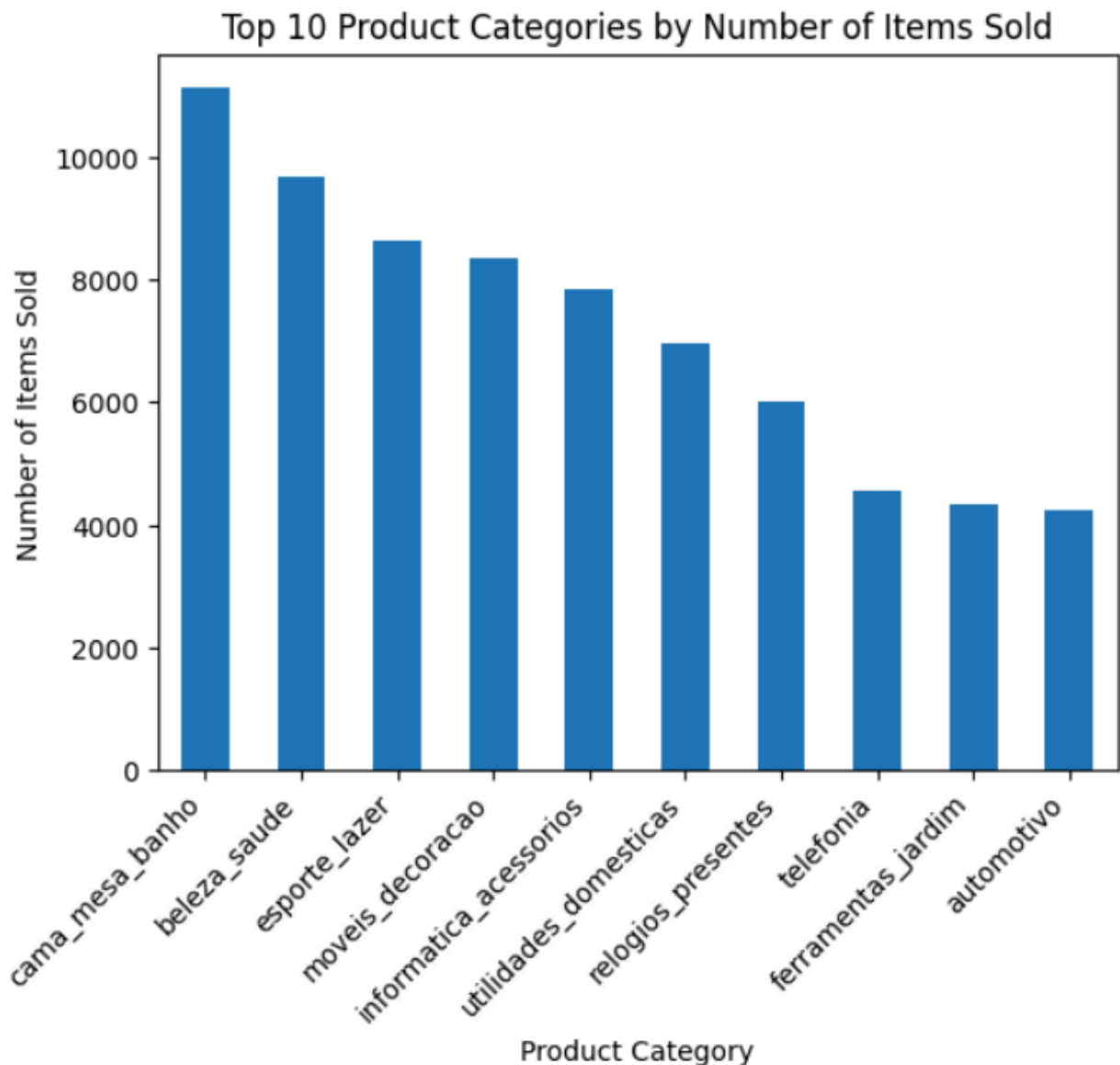
```
top_categories.plot(kind='bar')
```

#plot

```
plt.title('Top 10 Product Categories by Number of Items Sold')
```

```
plt.xlabel('Product Category')
```

```
plt.ylabel('Number of Items Sold')
plt.xticks(rotation=45, ha='right')
plt.show()
```



Explanation: The order-items dataset was combined with the products dataset using **product_id** so that each ordered item could be associated with its product category. The **value_counts()** function was then used on **product_category_name** to count the number of items sold in each category. The **head(10)** function selected the ten categories with the highest number of items. Finally, a horizontal bar chart was created to make the category names easier to read. The bar chart shows the top 10 product categories based on the number of items sold. Cama mesa banho is among the highest-selling categories, indicating strong customer demand for products in this category.

Task 6: Power BI Dashboard

The Power BI dashboard was created to provide an interactive view of the Olist e-commerce sales and order performance. KPI cards, charts, and slicers were used to monitor order volume, sales, product categories, order status, and payment methods.

1)3 KPI Cards.

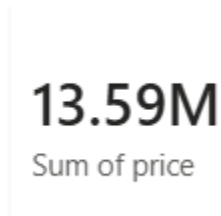
1. Total Orders:



Explanation:

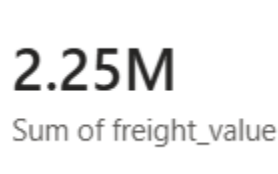
This KPI represents the total number of unique orders in the Olist e-commerce dataset. We use `DISTINCTCOUNT(order_id)` because each unique `order_id` represents one order. Power BI displays 99K as an abbreviation for approximately 99,000 orders.

2. Total Sales:



This KPI represents the total sales value of the products in the dataset. It is calculated by taking the sum of the price column from the order-items table. The total is approximately R\$13.59 million.

3. Total Freight:



This KPI represents the total freight/shipping charges associated with the order items. It is calculated using the sum of the `freight_value` column. The total freight is approximately R\$2.25 million.

2)4 different visualizations:

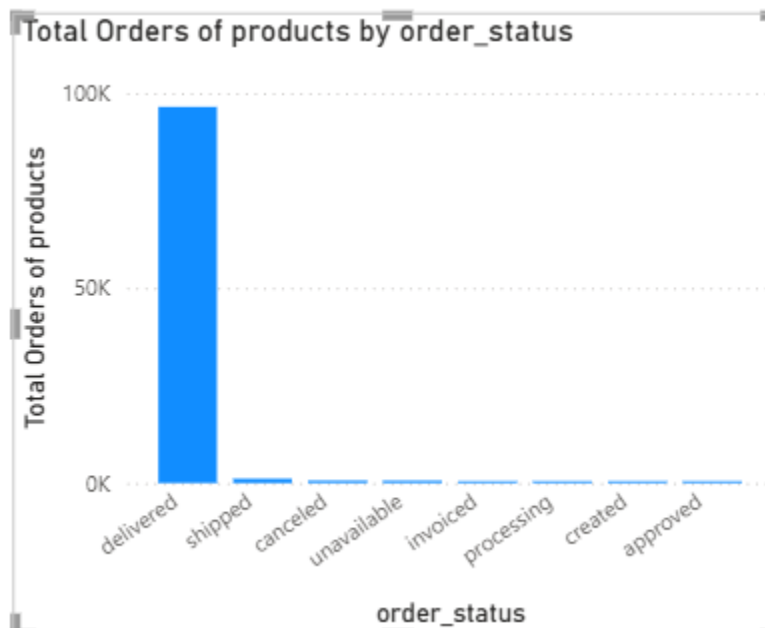
1.Total Orders of Products by Year-line chart



Explanation:

A line chart was used to analyze the total number of products ordered across different years. The chart shows a clear increase in order activity from 2016 to 2018, with the highest number of orders recorded in 2018. This indicates that the e-commerce business experienced growth in order activity over the period.

2.Order by status-Column chart

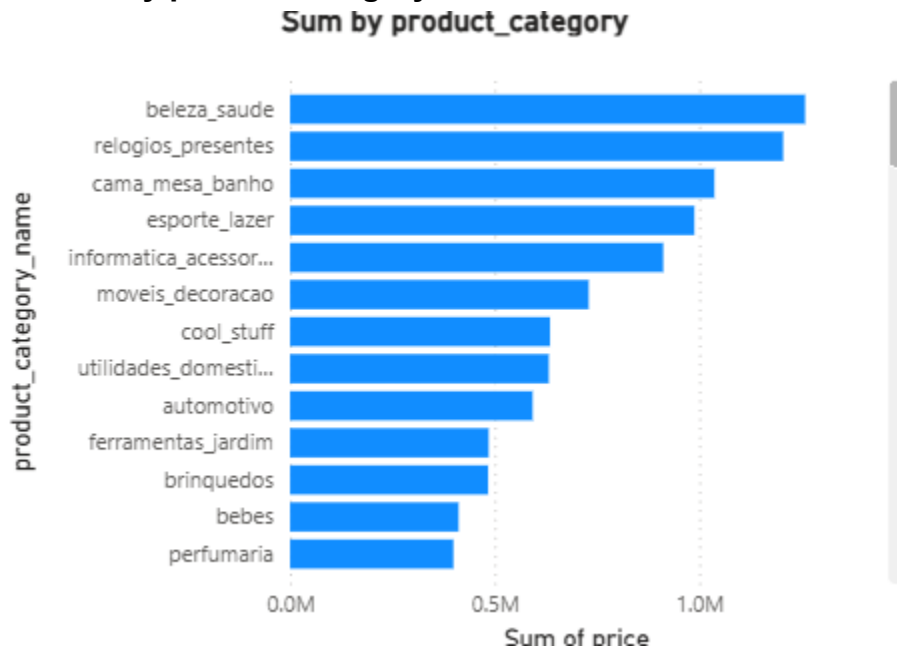


Explanation

A clustered column chart was used to compare orders according to their status ,such as delivered,cancelled,shipped, unavailable and other order stages.

Finding: Delivered orders are the dominated category, with approximately **96,478** **delievered** orders out of 99,441 total unique orders.This indicates the most orders successfully reached the delivered stage.

3.Sales by product category-Bar chart



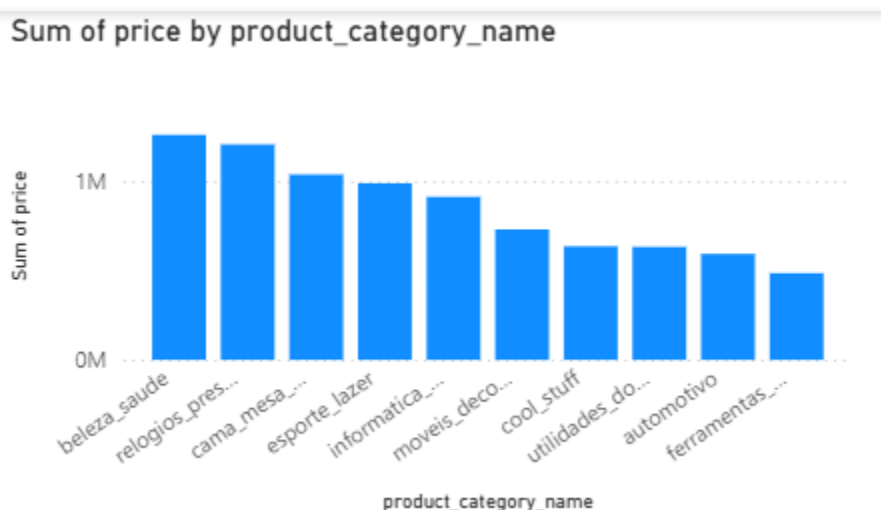
Explanation:

A bar chart was used to compare the total sales generated by different product categories.

The product category is displayed on one axis and sales value on the other. This helps identify which categories contribute more to overall sales.

Finding: Sales are concentrated in a smaller number of product categories, showing that some categories contribute significantly more to overall revenue than others.

4. Top 10 Product Categories by Sales — Bar Chart



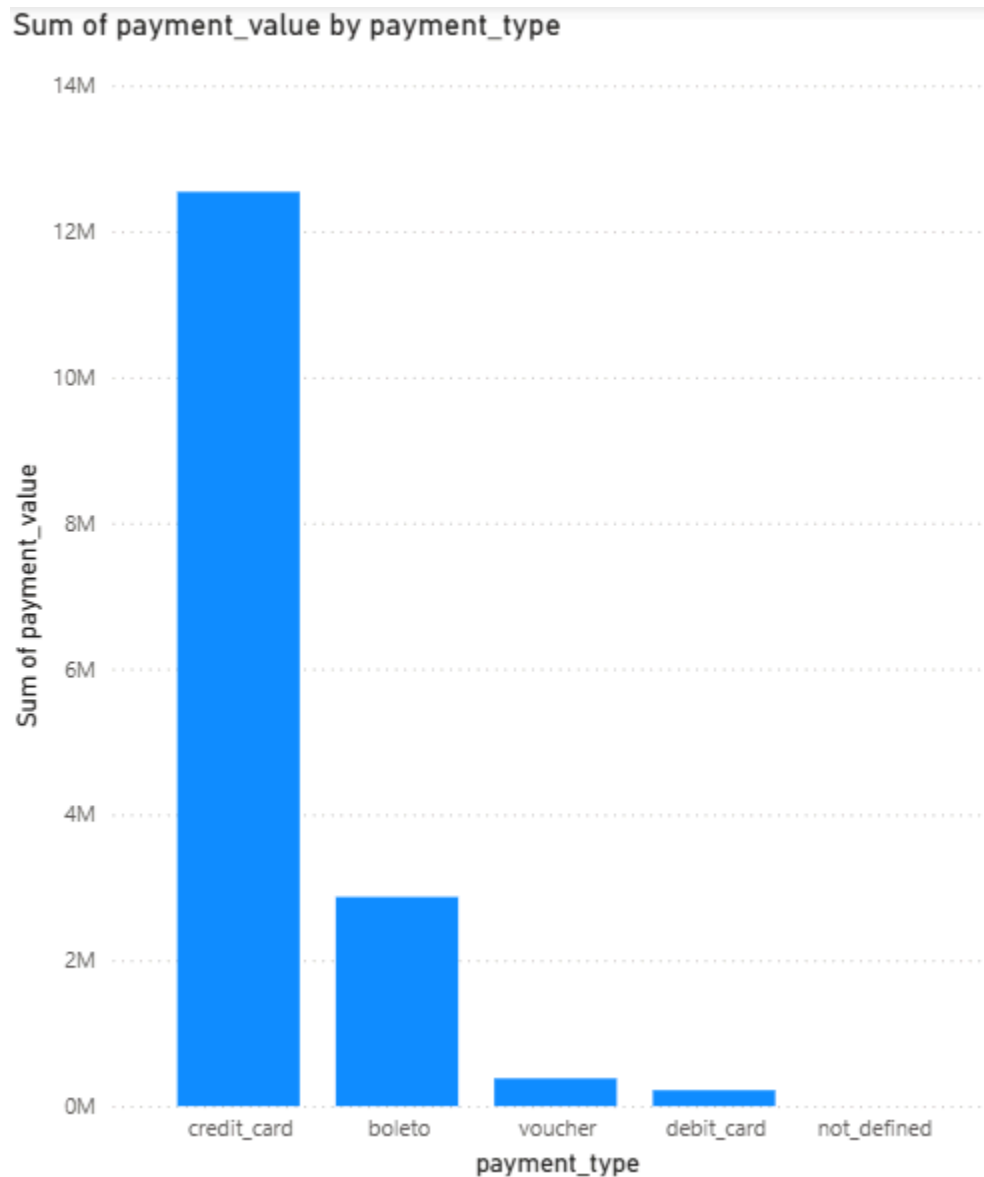
Explanation:

A Top 10 bar chart was created to focus specifically on the product categories generating the highest sales.

The Top N feature in Power BI was used to display only the **ten highest-performing categories**. This makes it easier to identify the major contributors to sales.

Finding: A relatively small group of product categories accounts for a significant portion of sales, which can help the business prioritize inventory and marketing activities.

5.Sales by Payment Type — Column Chart



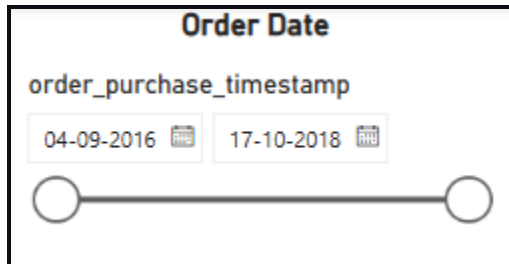
Explanation:

A column chart was created to compare sales according to different payment methods, such as credit card, boleto, voucher, and debit card.

Finding: Credit card is the leading payment type by payment value, indicating that customers predominantly use credit cards for their purchases.

3) Filters/Slicers:

1. Order Date

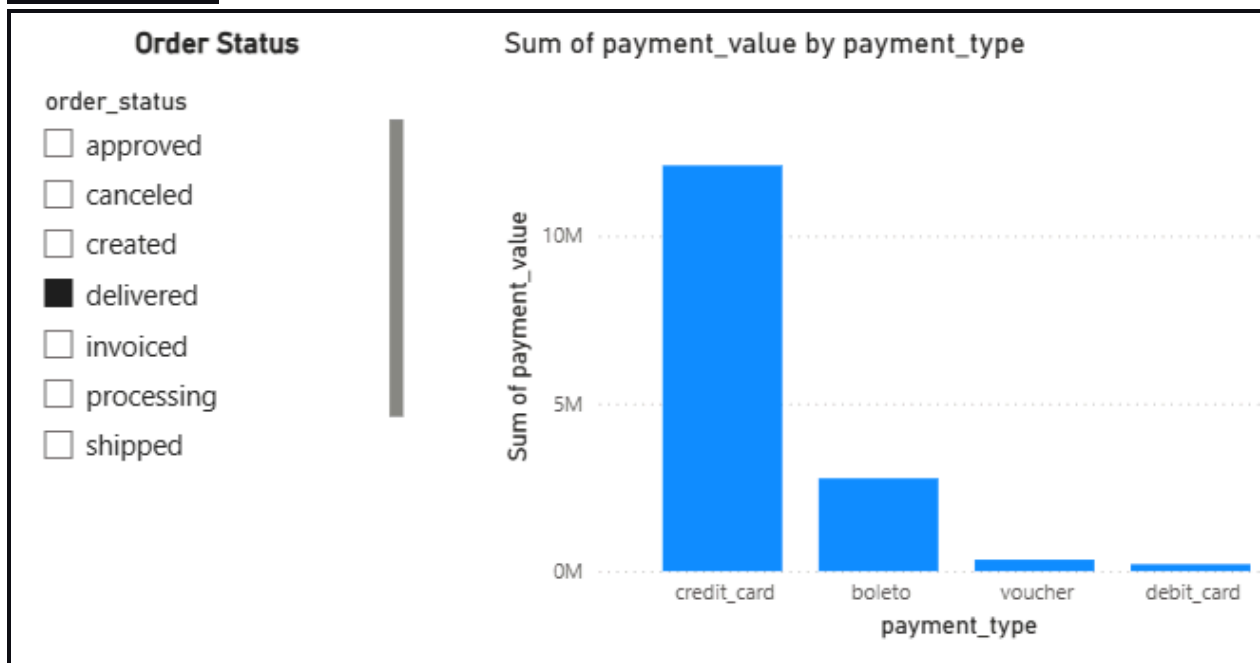


The image shows a date range slicer titled "Order Date". It features a label "order_purchase_timestamp" above two date input fields. The first field contains "04-09-2016" and the second contains "17-10-2018". Below these fields is a horizontal timeline with circular endpoints at each date, indicating a range selection.

Explanation:

An Order Date slicer was added to allow users to filter the dashboard based on the date range of orders. Users can select a specific date or a range of dates, and the other dashboard visuals update to show the corresponding order and sales information. This makes it easier to analyze e-commerce performance for a particular time period.

2. Order Status



Explanation: An Order Status slicer was added to allow users to filter the dashboard based on the status of orders, such as delivered, canceled, unavailable, and other order stages. Selecting a particular status updates the dashboard visuals to show the corresponding sales and order information. This helps analyze the performance of orders at different stages.

3. Product Category:

Product Category

product_category_name

☐

☐ agro_industria_e_comercio

☐ alimentos

☐ alimentos_bebidas

☒ artes

☐ artes_e_artesanato

☐ artigos_de_festas

☐ artigos_de_natal

☐ audio

☐ automotivo

☐ bebes

Explanation:

A Product Category slicer was added to allow users to filter the dashboard based on individual product categories. Users can select a category to view the corresponding orders and sales information across the dashboard. This makes it easier to analyze the performance of specific product categories.

4) Dashboard Title:

Olist E-commerce Sales & Order Performance Dashboard

5) Layout:

Olist E-Commerce Sales & Order Performance Dashboard

99K

Total Orders of products

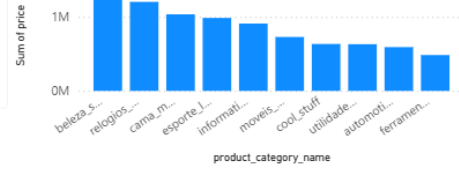
13.59M

Sum of price

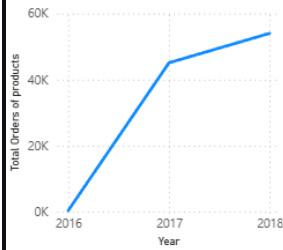
2.25M

Sum of freight_val...

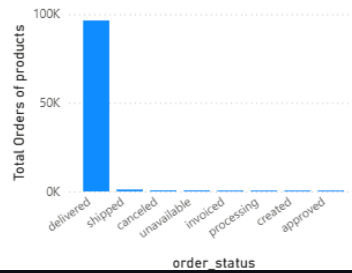
Sum of price by product_category_name



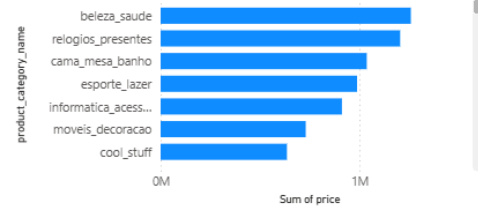
Total Orders of products by Year

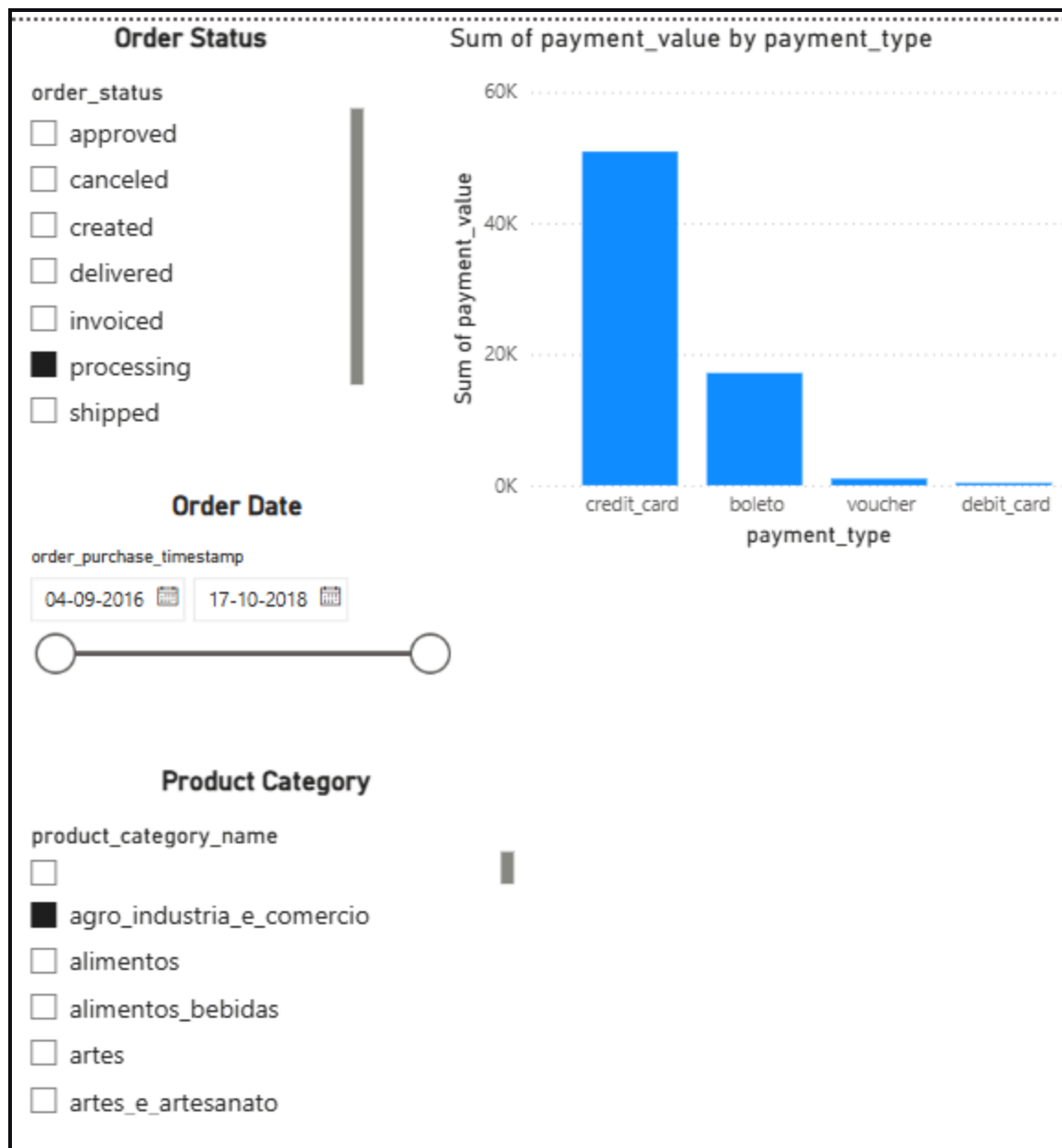


Total Orders of products by order_status



Sum by product_category





Task 7: Business Insights & Recommendations

1) Business Metrics:

1. Strong order completion performance

The dataset contains 99,441 unique orders.

Around 96,478 orders were successfully delivered, showing efficient order fulfillment and logistics performance.

2. High revenue generation

Total sales are approximately R\$13.59 million.

This indicates strong customer demand and healthy business growth.

3. Credit cards are the preferred payment method

The Credit Card payment type has the highest payment value.

Customers prefer digital and convenient payment methods.

4. Product prices show a wide range

Average item price: R\$120.65

Median item price: R\$74.99

Since the average is higher than the median, the data is right-skewed, meaning a few expensive products increase the average price.

5. Freight cost increases with product price

Correlation between price and freight value = 0.414.

Higher-priced products generally have higher shipping costs.

2) Explain important trends and patterns

Monthly orders increased noticeably from late 2017, showing business growth.

Most orders are in Delivered status, while canceled and unavailable orders are comparatively low.

Customers strongly prefer credit card payments.

Sales are concentrated in a few top-performing product categories.

Some products have extremely high prices (outliers), with prices reaching up to R\$6,735.

3) High / Low Performing Areas

✓ Delivered orders are very high.

✓ Credit card payment method performs best.

✓ Top product categories contribute significantly to sales.

Low Performing Areas

- Canceled and unavailable orders still exist.
- High freight costs may affect customer satisfaction.
- Some product categories contribute very little to revenue.

4) Data-Driven Business Recommendations

Recommendation 1:

Improve logistics and reduce freight costs.

Since freight increases with product price, optimizing delivery routes and shipping partnerships can reduce costs and improve customer satisfaction.

Recommendation 2:

Promote high-performing product categories and payment methods.

Focus marketing efforts on top-selling product categories.

Provide offers, discounts, and cashback for credit card payments to increase sales further.

Recommendation 3:

Reduce order cancellations.

Analyze reasons for canceled and unavailable orders to improve customer experience and operational efficiency.

Task 8:GitHub Documentation & Project Presentation

Olist E-Commerce Sales & Order Performance Analysis

1. Project Overview

This project focuses on analyzing the Olist Brazilian E-Commerce Public Dataset to understand e-commerce sales performance, customer orders, product categories, payment methods, order status, and delivery-related information.

The project combines Python-based data analysis and visualization with an interactive Power BI dashboard to identify important business insights and provide data-driven recommendations.

2. Problem Statement

E-commerce businesses generate large amounts of data related to orders, products, sales, payments, and delivery. Analyzing this data can help businesses understand sales performance, identify high-performing product categories, monitor order trends, and improve operational decision-making.

The objective of this project is to analyze the Olist e-commerce dataset and transform the raw data into meaningful insights using Python and Power BI.

3. Dataset Description

The project uses the Olist Brazilian E-Commerce Public Dataset.

The dataset contains approximately 100,000 orders from the Brazilian e-commerce marketplace and includes information about:

- Customers

- Orders

- Order items

- Products

- Sellers

- Payments

- Reviews

- Geolocation

- Product categories

The main datasets used for the analysis and Power BI dashboard were:

- "olist_orders_dataset"

- "olist_order_items_dataset"

- "olist_products_dataset"

- "olist_order_payments_dataset"

4. Tools Used

The following tools and technologies were used:

- Python – Data analysis and visualization
- Pandas – Data manipulation and cleaning
- Matplotlib – Data visualization
- Google Colab – Python development environment
- Power BI – Interactive dashboard and business analysis
- Git – Version control
- GitHub – Project repository and documentation

5. Data Cleaning Process

The dataset was inspected and prepared before analysis.

The following steps were performed:

- Loaded the required Olist CSV datasets
- Inspected the structure and columns of the datasets
- Identified the data types of important columns
- Checked for missing values
- Checked for duplicate records
- Converted relevant order-date columns into proper datetime format
- Retained missing delivery-related dates where appropriate because they represent orders that had not reached the corresponding stage of the delivery process
- Prepared the cleaned datasets for Exploratory Data Analysis and Power BI visualization

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the characteristics and patterns in the dataset.

The analysis included:

- Descriptive statistics
- Missing-value analysis
- Duplicate checking
- Product price analysis
- Freight analysis
- Correlation analysis
- Order-status analysis
- Monthly order trends
- Product-category analysis
- Outlier analysis

Important EDA Findings

- The average product price was approximately R\$120.65
- The median product price was approximately R\$74.99

- Product prices showed a right-skewed distribution, indicating that most products had lower prices while a smaller number of products had considerably higher prices.
- The correlation between product price and freight value was approximately 0.414, indicating a moderate positive relationship.
- Monthly order activity increased toward the later part of 2017.
- Some product categories contributed significantly more items and sales than others.
- High-price observations were identified through IQR-based outlier analysis. These observations were retained because they may represent genuine high-value products.

7. Data Visualizations

Python was used to create visualizations for understanding the dataset.

The following visualizations were created:

1. Distribution of Product Prices

A histogram was used to understand the distribution of product prices.

2. Product Price vs Freight Value

A scatter plot was used to examine the relationship between product price and freight value.

3. Orders by Status

A bar chart was used to compare different order statuses.

4. Monthly Number of Orders

A line chart was used to identify changes in order activity over time.

5. Top 10 Product Categories by Items Sold

A bar chart was used to identify the most frequently sold product categories.

The visualization files are available in the "visualizations" folder of this repository.

8. Power BI Dashboard

An interactive Power BI dashboard titled:

Olist E-Commerce Sales & Order Performance Dashboard

was developed to provide an overview of sales and order performance.

Key Performance Indicators

- Total Orders: 99,441 unique orders
- Total Sales: approximately R\$13.59 million
- Total Freight: approximately R\$2.25 million

Dashboard Visuals

The dashboard includes:

- Monthly Number of Orders
- Orders by Status
- Sales by Product Category
- Top 10 Product Categories by Sales
- Sales by Payment Type

The dashboard also includes **sliders** for:

- Order Date
- Order Status
- Product Category

The Power BI dashboard file is included in this repository

9. Key Insights

The major insights obtained from the analysis are:

1. The dataset contains 99,441 unique orders, with 96,478 orders delivered.
2. Total sales generated from the analyzed order-item data were approximately R\$13.59 million.
3. Total freight value was approximately R\$2.25 million.
4. Credit card payments represented the highest payment value among the payment types analyzed.
5. Product prices were right-skewed, with the mean price higher than the median price.
6. Product price and freight value showed a moderate positive correlation.
7. Order activity increased toward the later months of 2017.
8. Sales were concentrated in a smaller number of high-performing product categories.

10. Business Recommendations

Based on the analysis, the following recommendations can be made:

Improve Logistics

Analyze freight costs by product category, seller, location, and order characteristics to identify opportunities for reducing shipping expenses and improving delivery efficiency.

Focus on High-Performing Categories

Businesses should prioritize inventory planning and marketing efforts for product categories that generate higher sales and order volumes.

Leverage Payment Preferences

Since credit card payments represent a major share of payment value, businesses can consider offers such as cashback, discounts, and installment-based promotions to encourage purchases.

Monitor Order Performance

Order-status analysis can help identify cancellations, unavailable orders, and other operational issues that may affect customer satisfaction.

Use Data-Driven Inventory Planning

Historical sales trends can be used to improve demand forecasting and maintain appropriate inventory levels for popular product categories.

11. Conclusion

This project demonstrates how real-world e-commerce data can be transformed into meaningful business insights using Python and Power BI.

The analysis covered data preparation, exploratory data analysis, statistical investigation, visualization, and interactive dashboard development. The findings provide useful information about sales performance, product categories, payment methods, order trends, and logistics.

The project also demonstrates the use of Git and GitHub for version control, project organization, and professional documentation.

Overall, the analysis shows how data analytics can support better business decisions in the e-commerce domain.

12. Project Files

The repository contains the following project resources:

- "README.md" – Project documentation
- "Final_Data_Analytics_project.ipynb" – Python analysis notebook
- Power BI dashboard file
- "practice.txt" – Git workflow practice file
- "visualizations/" – Python and Power BI visualization screenshots

Project Workflow

Data Collection → Data Inspection → Data Cleaning → EDA → Visualization → Power BI Dashboard → Insights → Business Recommendations